Source A · NHS public information

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Cystic fibrosis is inherited. Altered gene copies from both parents cause thick, sticky mucus. This can affect the lungs and digestion. Treatments can help ease symptoms. Care depends on the person.

This source gives a public overview. It does not tell us one person’s exact symptoms or future.

Publisher: NHS. Title: Cystic fibrosis. Use the Causes, Symptoms and Treatments sections.

Source A full page: ask an adult to use the link in the page tools or staff notes.

Accessed 24 October 2026. Page reviewed 28 February 2025. This is a classroom paraphrase, not a quotation. The full page can be opened by the teacher.

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Source B · NHS Genomics Education

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Cystic fibrosis usually involves harmful variants in both copies of CFTR. This is an autosomal recessive condition. A carrier has one changed copy in this model. The effects can vary.

Two carrier parents: each pregnancy	Model chance
Child has the condition	1 in 4 = 25%
Child is a carrier	2 in 4 = 50%
Child inherits neither changed copy	1 in 4 = 25%

These chances apply each time. They do not predict a set family count or the severity of one person’s symptoms.

NHS Genomics Education Programme. Page: Cystic fibrosis. Read Genomics, then Inheritance and genomic counselling. Use these parts to check the gene and how changed copies pass to a child.

Source B full page: ask an adult to use the link in the page tools or staff notes.

Checked 24 October 2026. This card uses our own words. It is not a quote. We have not given a review date for this source.

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Sources · purpose and limits

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Both sources are NHS sources. They have different readers and are not wholly independent publishers.

Source A	Source B
Public information about effects and care	More detail about genes and inheritance

Both support inherited cause. More detail does not by itself mean a conflict. Check the actual claim, source section, date and purpose.

Read the source, cover it, explain the meaning in your own words, then check it. Keep the source label beside each claim.

Use fictional cases. Do not ask for family details or predict a person’s future. Inheritance and infection precautions are separate matters.

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Arrival · Independent

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Focus

What are the two main parts of the research question?

Help: Inheritance and a biological effect.

2. Source

How can your audience check a claim?

Help: Make the source visible.

3. Uncertainty

What can you say when the sources do not answer a question?

Help: An honest gap is allowed.

4. Choice

Must this be a public speech?

Help: You may show your work to one person or write it down.

My response: _____

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Independent · Independent

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Create

Complete the four-panel planning sheet.

Help: Use one accurate claim per panel rather than long paragraphs.

2. Present

Communicate the explanation using your chosen mode.

Help: A private one-to-one explanation is equally valid.

3. Respond

Answer: “Will every person with CF have exactly the same symptoms?”

Help: Refer to variation and the limits of the sources.

My response: _____

One partner explains. The other asks for the clue. Swap roles. Keep your own first answer, feedback and next step.

Use one route for the full 10 minutes. Keep your own record and response.

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Exit · Independent

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

1. Explain

Use two sentences. Explain how CF is passed on through genes. Link this to one effect on the body.

Help: Use CFTR, two altered alleles and thick mucus.

2. Qualify

State one thing that your sources do not show. Keep this limit in your talk or work.

Help: The source does not show one person’s future. Say what it does show.

My response: _____

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Four panels · planning sheet

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Plan one claim for each panel. Show the source for each claim. Use two named sources. Keep one limit. You may show it to your partner, write or direct a scribe.

Panel	My own explanation
1. A bounded question	_____
2. Recessive inheritance model	_____
3. One effect and its source	_____
4. Two sources and one uncertainty	_____

My chosen mode: _____

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Audience question and my response

Present inheritance and one biological effect.

Connect claims to two visible source labels.

Answer a question without exceeding the evidence.

Audience question: will every person with CF have exactly the same symptoms?

What my sources support: _____

What they do not establish: _____

A responsible next check: _____

What the adult heard: _____

My next response, or why I kept it: _____

Keep your first answer. Hear what the adult noticed. Make one real change, or give a reason to keep it. Ask for a read-back if it helps.

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